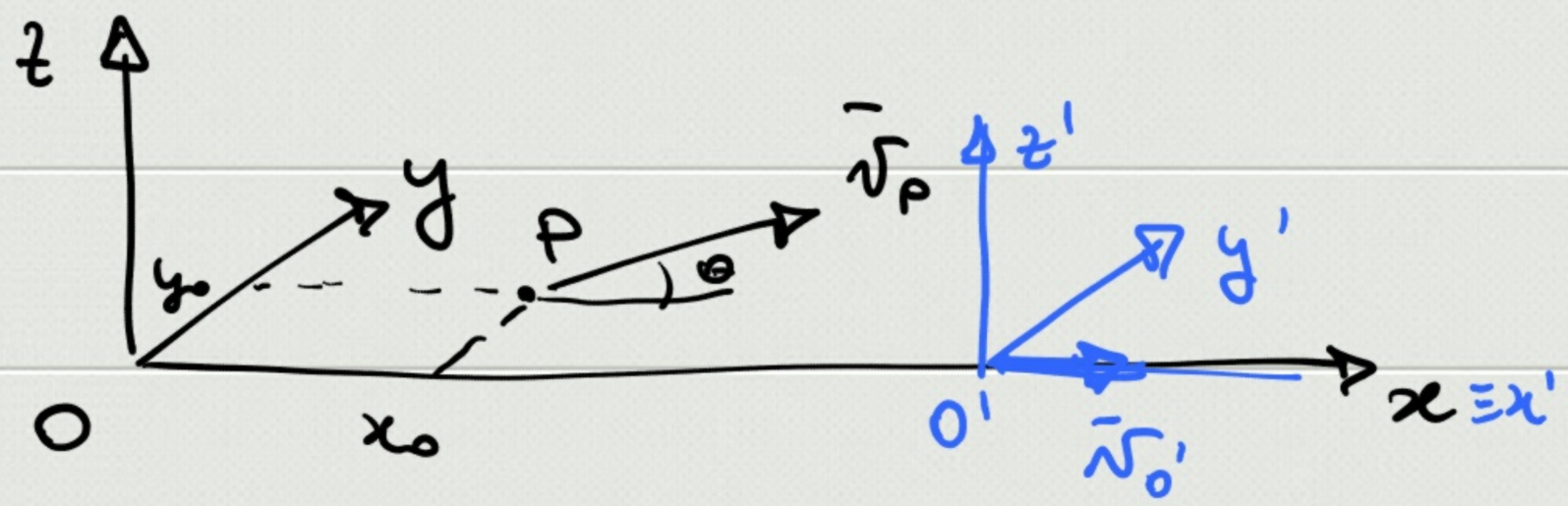
$$\begin{cases} \bar{r}' = \bar{r} - \bar{r}_{o'} & * \\ \bar{v}' = \bar{v} - \bar{v}_{o'} - \bar{\omega} \times \bar{r}' & \Leftarrow \\ \bar{a}' = \bar{a} - \bar{a}_{o'} - \bar{\omega} \times (\bar{\omega} \times \bar{r}') - \dfrac{d\bar{\omega}}{dt} \times \bar{r}' - 2\,\bar{\omega} \times \bar{v}' \end{cases}$$



$P = (x_0, y_0, 0)$

$\bar{v}_P = (v_x, v_y, 0)$

$\doteq$ costante

$$v = \sqrt{v_x^2 + v_y^2} \qquad \theta = \operatorname{atg}\left(\frac{v_y}{v_x}\right)$$

$$\bar{r}(t) = \bar{r}_0 + \bar{v}\,t \quad \Rightarrow \quad \begin{cases} x(t) = x_0 + v_x t \\ y(t) = y_0 + v_y t \end{cases}$$

$t = 0 \; : \quad O \equiv O'$

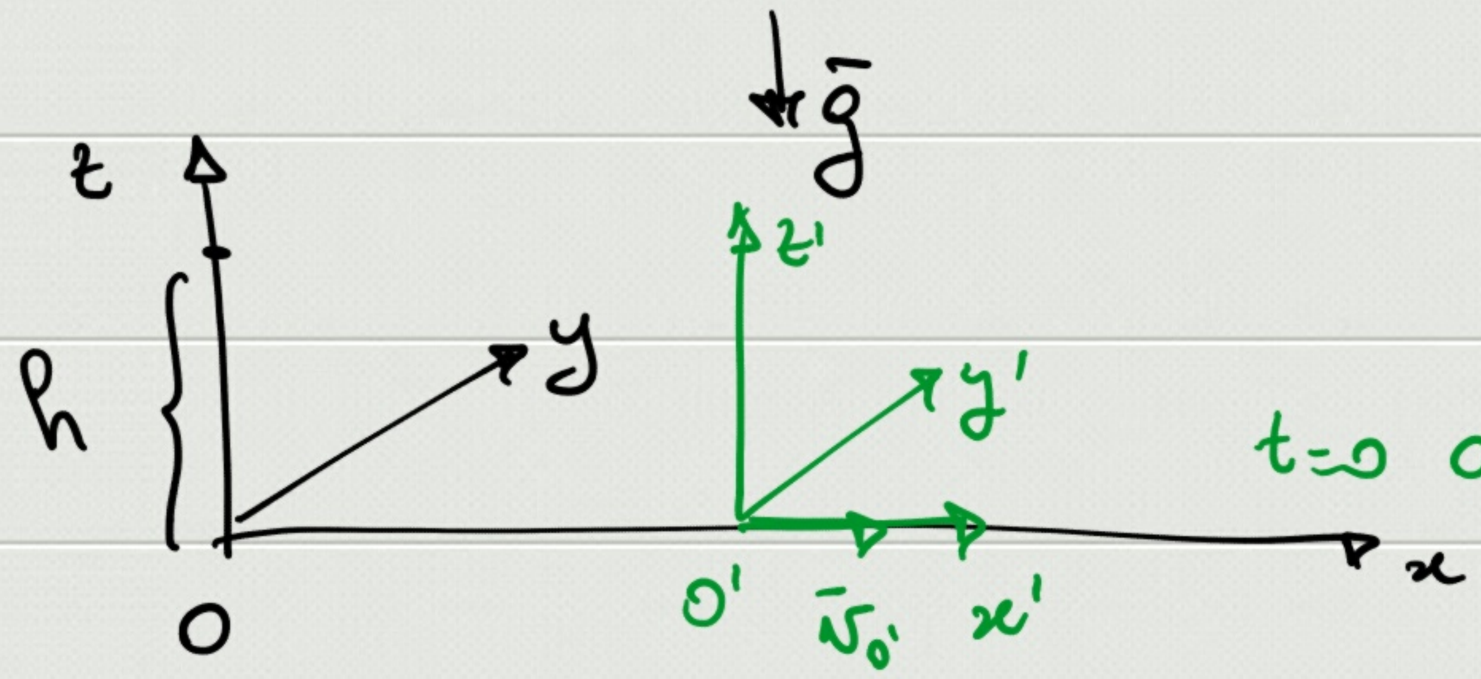
$$\bar{v}_{o'} = v_{o'}\,\bar{u}_x$$

$$x_{o'}(t) = v_{o'} t \quad \Leftarrow \quad \bar{r}_{o'}$$

$$\bar{r}' \doteq \bar{r} - \bar{r}_{o'} = \begin{cases} x'(t) = (x_0 + v_x t) - v_{o'} t = x_0 + (v_x - v_{o'})t \\ y'(t) = (y_0 + v_y t) \end{cases}$$

$$\bar{v}' = (v_x - v_{o'})\,\bar{u}_x + v_y\,\bar{u}_y$$

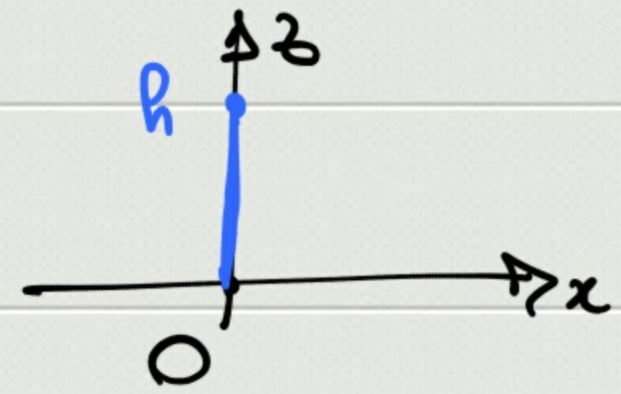
$$v' = \sqrt{(v_x - v_{o'})^2 + v_y^2} \neq v \qquad \theta' = \operatorname{atg} \frac{v_y}{v_x - v_{o'}}$$



$v_0 = 0$

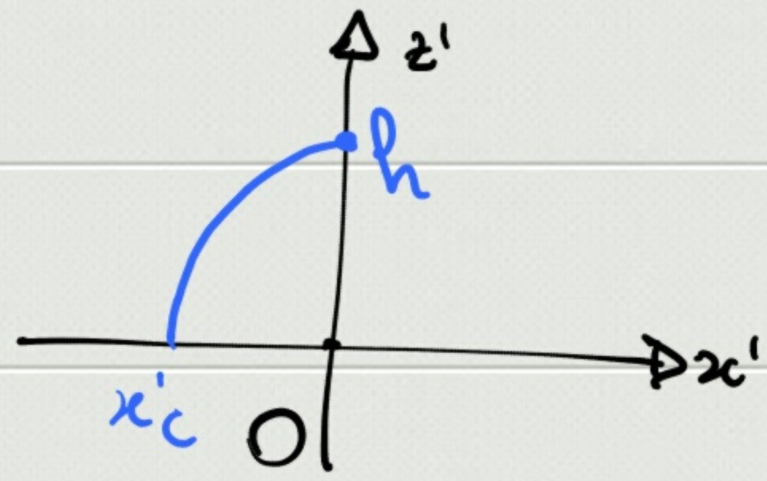
$t=0 \quad O \equiv O' \qquad \bar{v}_{O'} = v_{O'}\,\bar{u}_x$

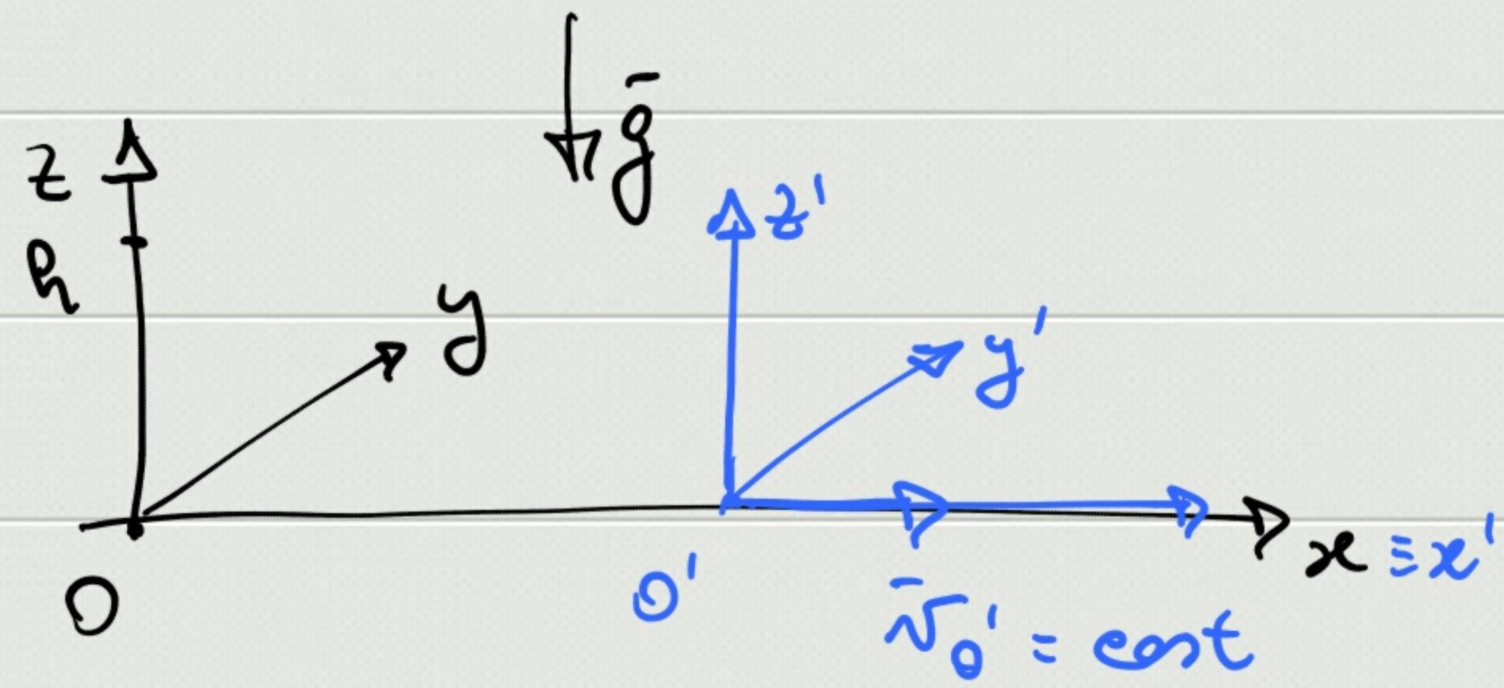
$$z(t) = h - \frac{1}{2} g t^2$$



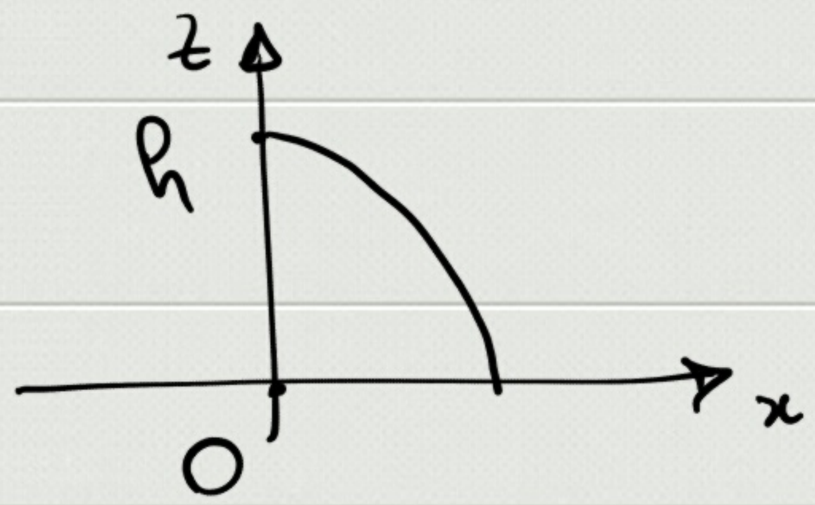
$\bar{r}' = \bar{r} - \bar{r}_{O'} \qquad \bar{r}_{O'} \Rightarrow x_{O'} = v_{O'} t$

$$\Rightarrow \begin{cases} x'(t) = -v_{O'} t \\ z'(t) = h - \frac{1}{2} g t^2 \end{cases}$$

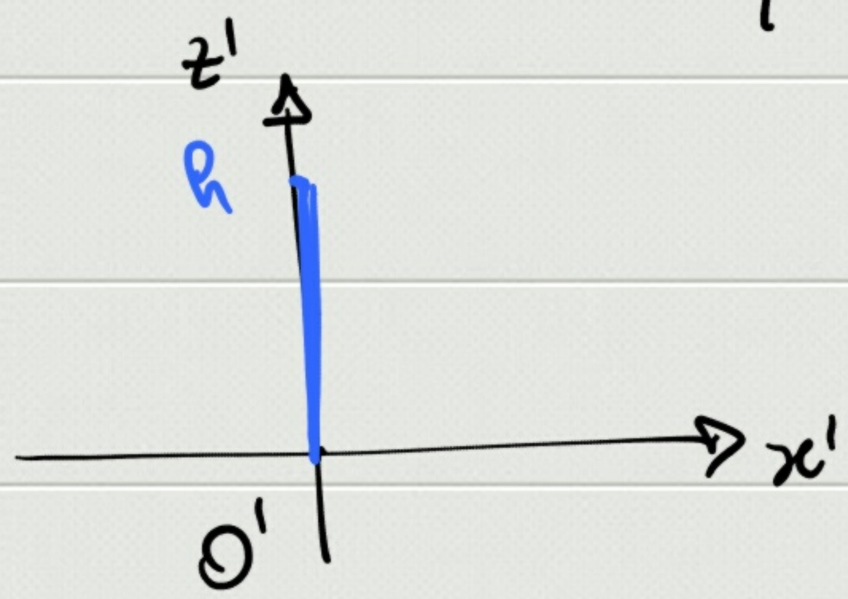


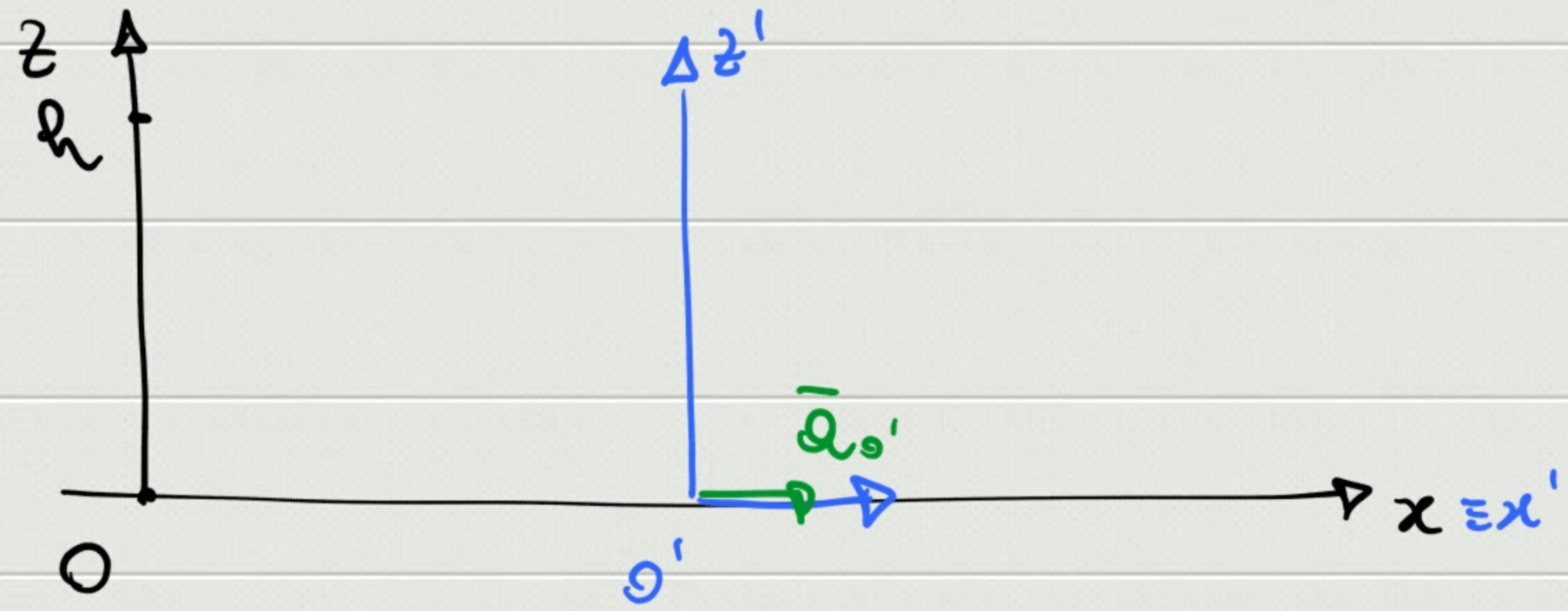


$$\bar{r}(t) = \bar{r}_0 + \bar{v}_0 t + \frac{1}{2}\bar{a}t^2 \Rightarrow \begin{cases} x(t) = v_{0'} t \\ z(t) = h - \frac{1}{2} g t^2 \end{cases}$$



$$\bar{r}'(t) = \bar{r} - \bar{r}_{O'} \Rightarrow \begin{cases} x'(t) = v_{0'} t - v_{0'} t = 0 \\ z'(t) = h - \frac{1}{2} g t^2 \end{cases}$$

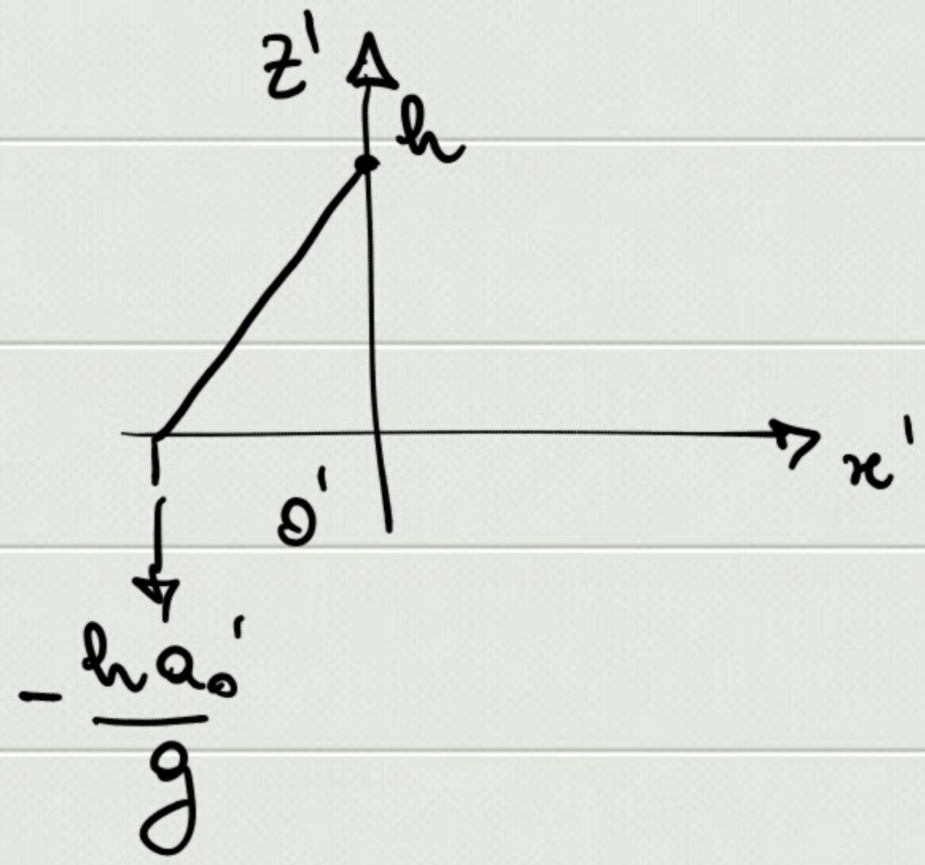
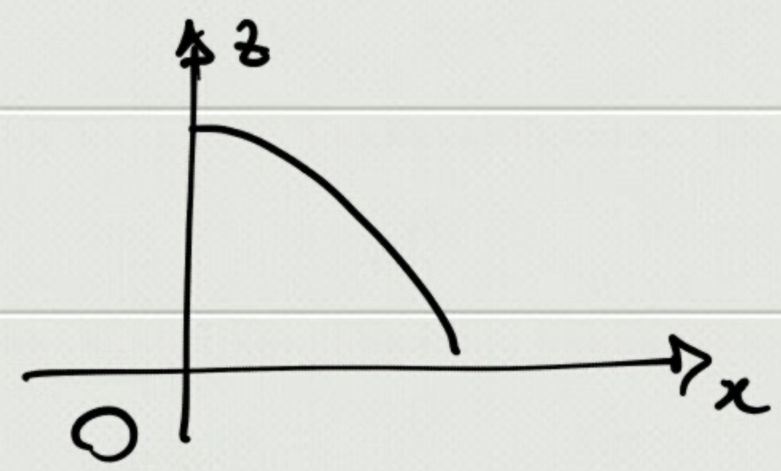




$$\bar{r}_{o'} = \emptyset + \bar{v}_{o',\emptyset}\, t + \frac{1}{2}\bar{a}_{o'} t^2$$

$$\Rightarrow \quad x_{o'}(t) = v_{o',\emptyset}\, t + \frac{1}{2} a_{o'} t^2$$

$$\bar{r}(t) = \bar{r}_o + \bar{v}_o t + \frac{1}{2}\bar{a} t^2 \Rightarrow \begin{cases} x(t) = v_{o',\emptyset}\, t \\ z(t) = h - \frac{1}{2} g t^2 \end{cases} \quad (*)$$

$$\bar{r}'(t) = \bar{r}(t) - \bar{r}_{o'}(t) \Rightarrow \begin{cases} x'(t) = \cancel{v_{o',\emptyset}\, t} - \cancel{v_{o',\emptyset}\, t} - \frac{1}{2} a_{o'} t^2 \\ z'(t) = h - \frac{1}{2} g t^2 \end{cases}$$

$$t^2 = -\frac{2x'}{a_{o'}} \quad \Rightarrow \quad z'(x') = h + \frac{1}{\cancel{2}} g \frac{\cancel{2}x'}{a_{o'}} = h + \frac{g}{a_{o'}} x'$$